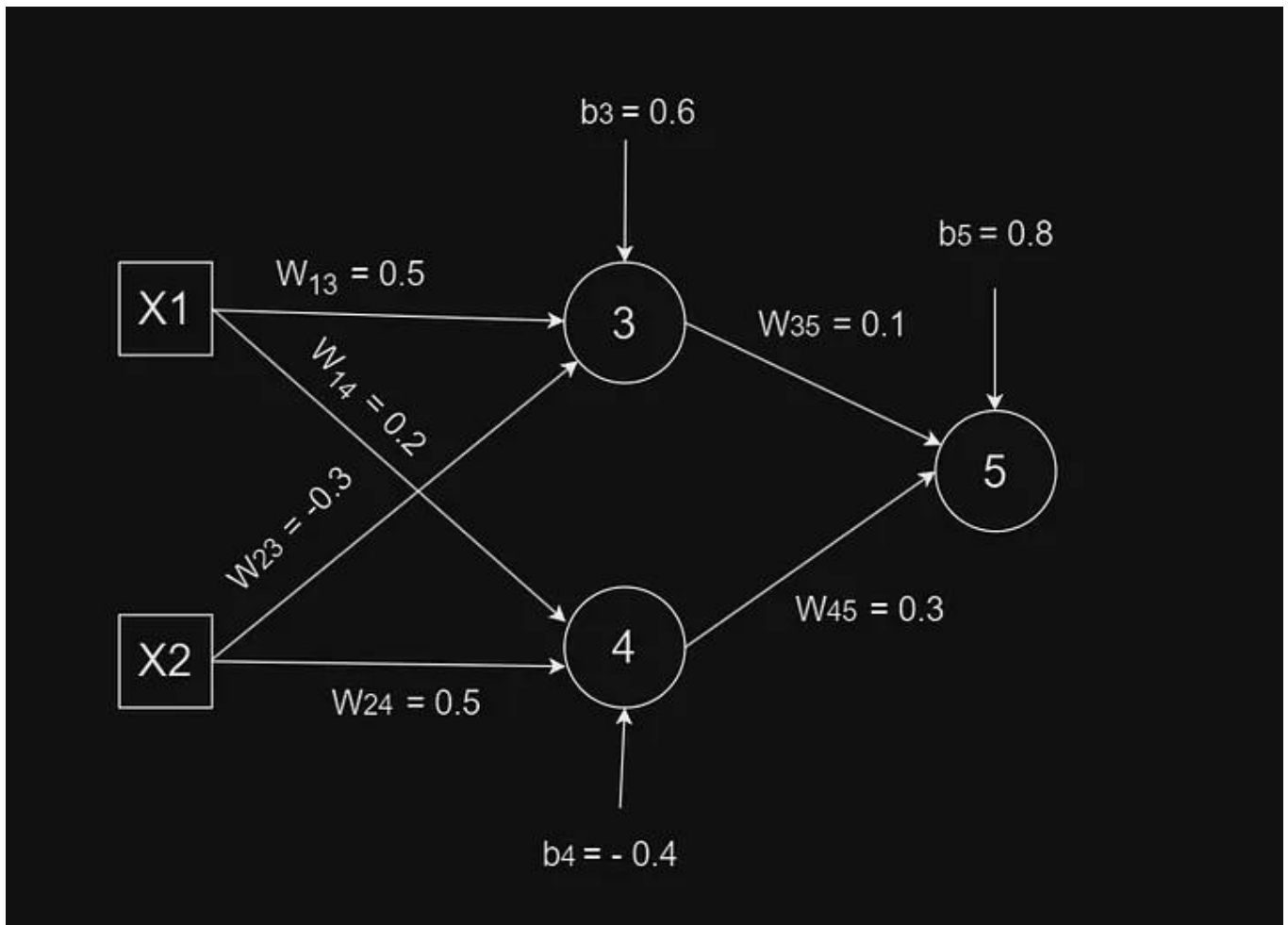


Artificial Neural Network

Q. Consider a *multilayer feed-forward neural network* given below. Let the *learning rate* be 0.5. Assume initial values of weights and biases as given in the table below. Train the network for the training tuples $(1, 1, 0)$ and $(0, 1, 1)$, where last number is target output. Show weight and bias updates by using back-propagation algorithm. Assume that sigmoid activation function is used in the network.

W13	W14	W23	W24	W35	W45	b3	b4	b5
0.5	0.2	-0.3	0.5	0.1	0.3	0.6	-0.4	0.8

Weight and Bias values



Neural Network

Artificial Neural Network

Solution:

Iteration 1

Input and Output values:

X1	X2	Output(τ)
1	1	0

Input and output for Iteration

a. Learning rate (l) = 0.5

b. Activation function = Sigmoid function

$$\text{(i.e. } \sigma = \frac{1}{1 + e^{-x}} \text{)}$$

Sigmoid Function

Weight(W) and bias(b) values:

W13	W14	W23	W24	W35	W45	b3	b4	b5
0.5	0.2	-0.3	0.5	0.1	0.3	0.6	-0.4	0.8

Weight and bias values for iteration 1

a. Calculate the input(I) and output(O) for each layer

-For input layer

$$O_1 = I_1 = 1$$

$$O_2 = I_2 = 1$$

Input and outputs of Input layer

-For hidden layer

$$\begin{aligned} I_3 &= O_1 \times W_{13} + O_2 \times W_{23} + b_3 \\ &= 1 \times 0.5 + 1 \times (-0.3) + 0.6 \\ &= 0.5 - 0.3 + 0.6 \\ &= 0.8 \end{aligned}$$

$$O_3 = \frac{1}{1 + e^{-I_3}} = \frac{1}{1 + e^{-0.8}} = 0.69$$

$$\begin{aligned} I_4 &= O_1 \times W_{14} + O_2 \times W_{24} + b_4 \\ &= 1 \times 0.2 + 1 \times 0.5 + (-0.4) \\ &= 0.2 + 0.5 - 0.4 \\ &= 0.3 \end{aligned}$$

$$O_4 = \frac{1}{1 + e^{-I_4}} = \frac{1}{1 + e^{-0.3}} = 0.5744$$

Inputs and outputs for hidden layer

-For output layer

$$\begin{aligned} I_5 &= O_3 \times W_{35} + O_4 \times W_{45} + b_5 \\ &= 0.69 \times 0.1 + 0.5744 \times 0.3 + 0.8 \\ &= 0.0690 + 0.1723 + 0.8 \\ &= 1.0413 \end{aligned}$$

$$O_5 = \frac{1}{1 + e^{-I_5}} = \frac{1}{1 + e^{-1.0413}} = 0.7391$$

b. Calculation of error at each node:

$$Err_j = O_j(1 - O_j)(T_j - O_j)$$

Error calculation formula for output layer

- **For output layer:**

$$\begin{aligned} Err_5 &= O_5 (1 - O_5) (\tau - O_5) \\ &= 0.7391 (1 - 0.7391) (0 - 0.7391) \\ &= 0.7391 (0.2609) (- 0.7391) \\ &= - 0.1425 \end{aligned}$$

Error calculation for output layer

$$Err_j = O_j(1 - O_j) \sum_k Err_k w_{jk}$$

Error calculation formula for hidden layer

- ***For hidden layer:***

$$\begin{aligned}\text{Err}_4 &= O_4 (1 - O_4) (\text{Err}_5 \times W_{45}) \\ &= 0.5744 (1 - 0.5744) (-0.1425 \times 0.3) \\ &= 0.5744 (0.4256) (-0.04275) \\ &= -0.010451\end{aligned}$$

$$\begin{aligned}\text{Err}_3 &= O_3 (1 - O_3) (\text{Err}_5 \times W_{35}) \\ &= 0.69 (1 - 0.69) (-0.1425 \times 0.1) \\ &= 0.69 (0.31) (-0.1425) \\ &= -0.030481\end{aligned}$$

c. Update the weight

$$W_{ij}(\text{new}) = W_{ij}(\text{old}) + I \times O_i \times \text{Err}_j$$

Formula for new weight calculation

$$W_{45}(\text{new}) = W_{45}(\text{old}) + I \times O_4 \times \text{Err}_5 = 0.3 + 0.5 \times 0.5744 \times (-0.1425) = 0.2591$$

$$W_{35}(\text{new}) = W_{35}(\text{old}) + I \times O_3 \times \text{Err}_5 = 0.1 + 0.5 \times 0.69 \times (-0.1425) = 0.0508$$

$$W_{24}(\text{new}) = W_{24}(\text{old}) + I \times O_2 \times \text{Err}_4 = 0.5 + 0.5 \times 1 \times (-0.010451) = 0.494775$$

$$W_{14}(\text{new}) = W_{14}(\text{old}) + I \times O_1 \times \text{Err}_4 = 0.2 + 0.5 \times 1 \times (-0.010451) = 0.194774$$

$$W_{23}(\text{new}) = W_{23}(\text{old}) + I \times O_2 \times \text{Err}_3 = -0.3 + 0.5 \times 1 \times (-0.030481) = -0.315240$$

$$W_{13}(\text{new}) = W_{13}(\text{old}) + I \times O_1 \times \text{Err}_3 = 0.5 + 0.5 \times 1 \times (-0.030481) = 0.484759$$

Updated weight values

Artificial Neural Network

d. Update the bias value

$$b_j = b_{j(\text{old})} + I \times \text{Err}_j$$

Formula for new bias calculation

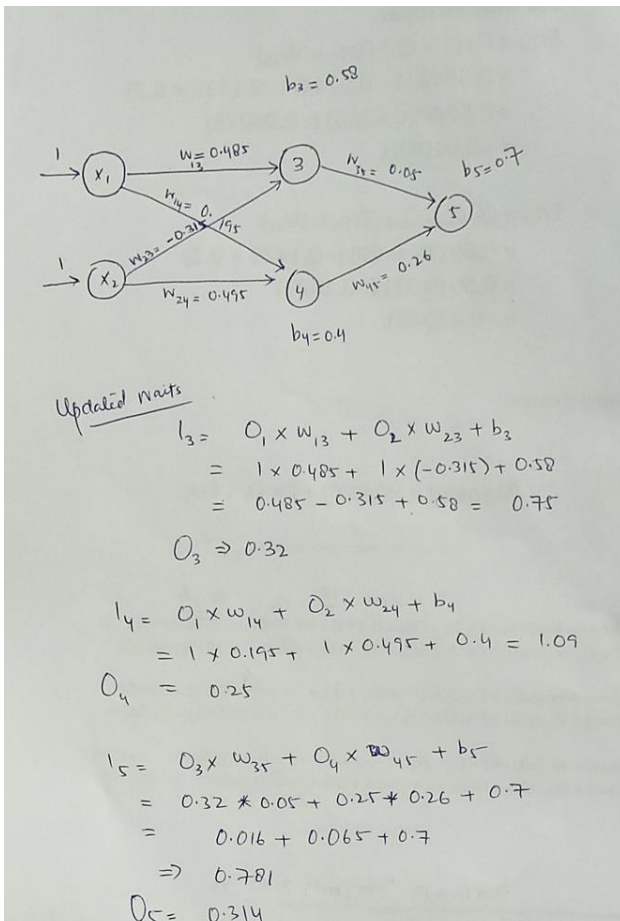
$$b_3 = b_{3(\text{old})} + I \times \text{Err}_3 = 0.6 + 0.5 \times (-0.030481) = 0.584759$$

$$b_4 = b_{4(\text{old})} + I \times \text{Err}_4 = -0.4 + 0.5 \times (-0.010451) = -0.405225$$

$$b_5 = b_{5(\text{old})} + I \times \text{Err}_5 = 0.8 + 0.5 \times (-0.1425) = 0.7288$$

Updated bias values

e. Again feed forward by updated weights and calculate error at O_5 . $O_5 = 0.314$ has less value comparatively than previous iteration that was 0.7391. our desired value is 0. Err_5 previously was -0.1425 after updating weights and biases it is now -0.067.



$$\begin{aligned} \text{Error at } O_5 &= O_5 (1 - O_5) (T - O_5) \\ &= 0.314 (1 - 0.314) (0 - 0.314) \\ &= 0.314 (0.686) (-0.314) \\ &= -0.067 \end{aligned}$$